

Calculation of Percentiles

- On Next Slide

Steps in Determining the Location of a Percentile

1. Organize the numbers into an ascending-order array.
2. Calculate the percentile location (i) by:

$$i = \frac{P}{100}(N)$$

where

P = the percentile of interest

i = percentile location

N = number in the data set

3. Determine the location by either (a) or (b).
 - a. If i is a whole number, the P th percentile is the average of the value at the i th location and the value at the $(i + 1)$ st location.
 - b. If i is not a whole number, the P th percentile value is located at the whole number part of $i + 1$.

For example, suppose you want to determine the 80th percentile of 1240 numbers. P is 80 and N is 1240. First, order the numbers from lowest to highest. Next, calculate the location of the 80th percentile.

$$i = \frac{80}{100}(1240) = 992$$

Because $i = 992$ is a whole number, follow the directions in step 3(a). The 80th percentile is the average of the 992nd number and the 993rd number.

$$P_{80} = \frac{(992\text{nd number} + 993\text{rd number})}{2}$$

Boxplots

Why use boxplots?

- ease of construction
- convenient handling of outliers
- construction is not subjective (like histograms)
- Used with medium or large size data sets ($n \geq 10$)
- useful for comparative displays

Disadvantage of boxplots

- does not retain the individual observations
- should **not** be used with small data sets ($n < 10$)

How to construct

- Find five-number summary

Min Q1 Med Q3 Max

- draw box from Q1 to Q3
- draw median inside the box

Determine inner and outer fences

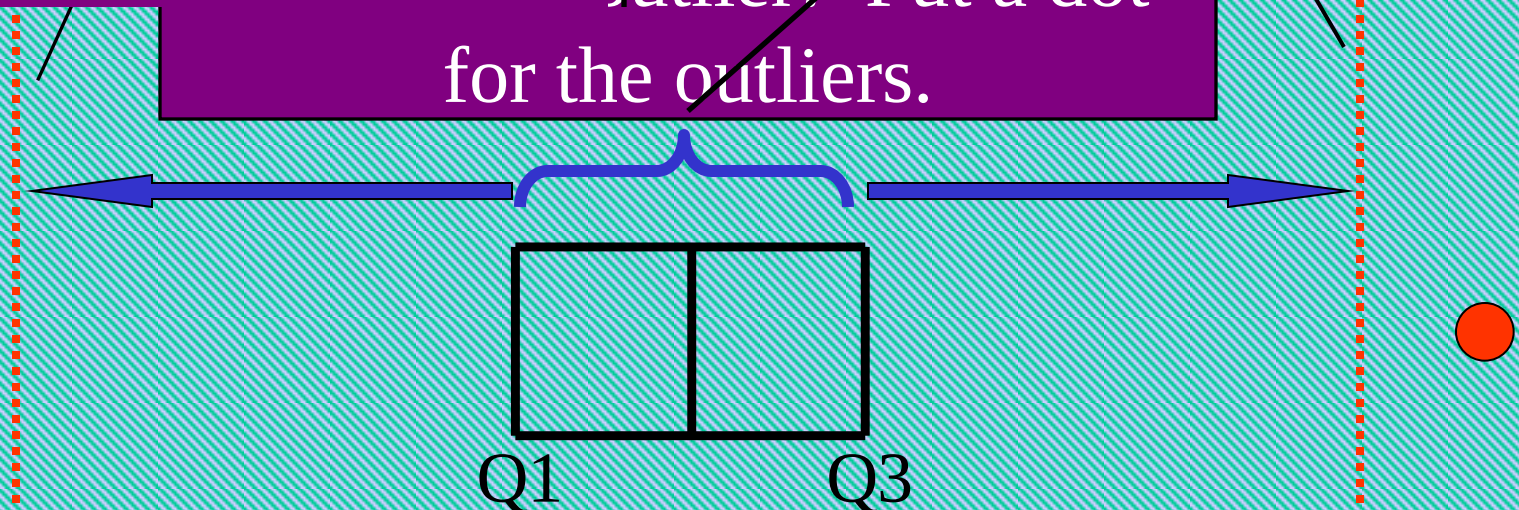
- The following quantities (called *fences*) are needed for identifying extreme values in the tails of the distribution
- lower inner fence: $Q1 - 1.5 * IQR$
- upper inner fence: $Q3 + 1.5 * IQR$
- lower outer fence: $Q1 - 3 * IQR$
- upper outer fence: $Q3 + 3 * IQR$

Inner fence

Interquartile Range (IQR) – is the range (length) of the box
 $Q3 - Q1$

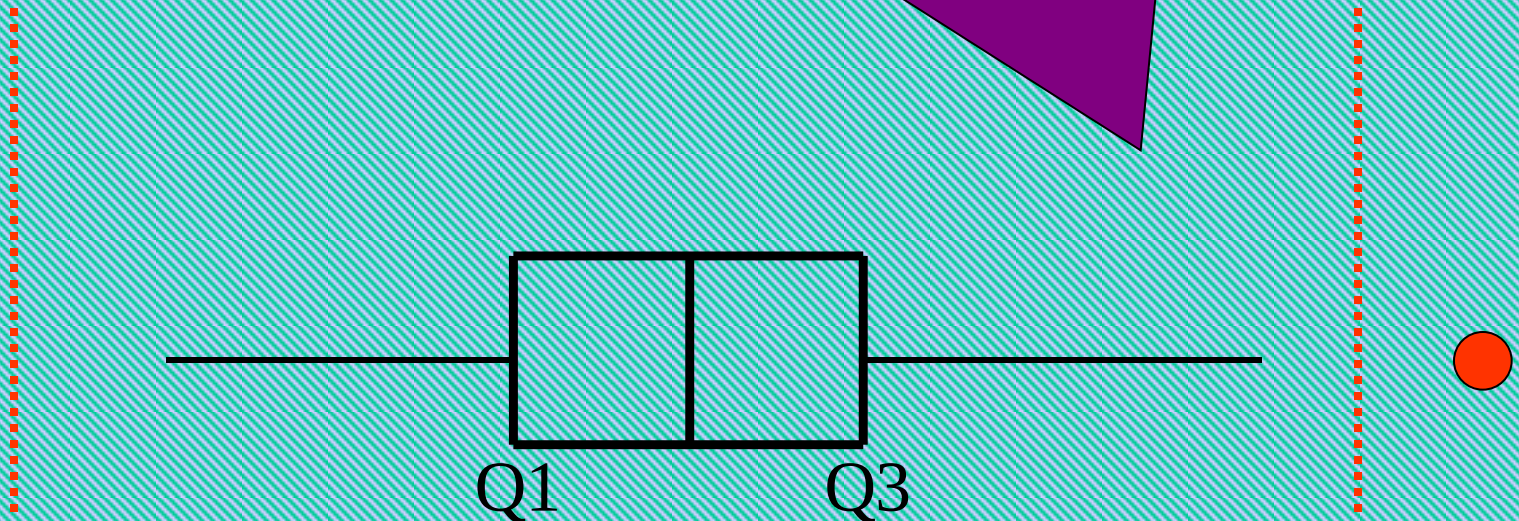
$$Q3 + 1.5IQR$$

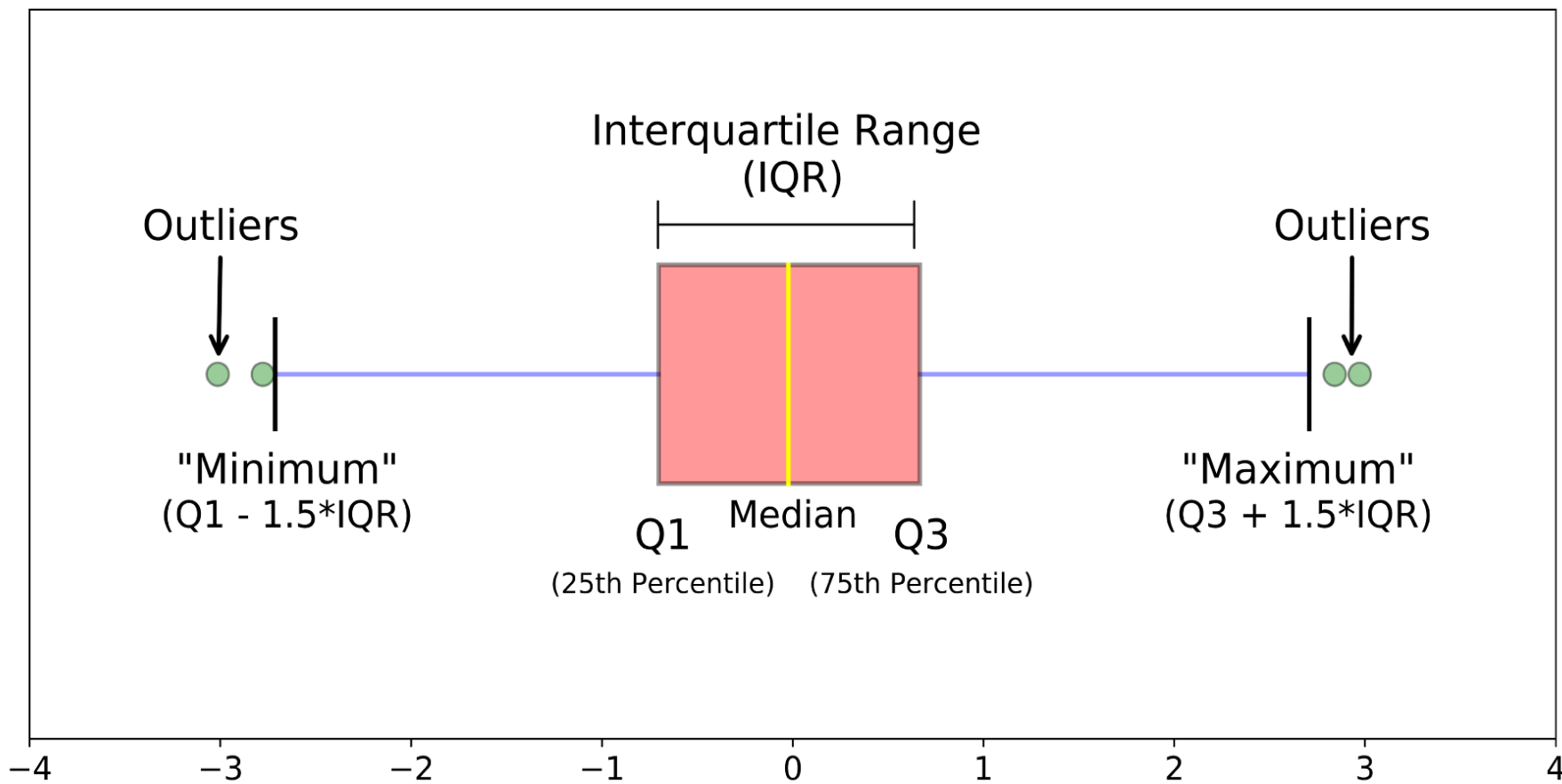
Information outside this box is an outlier! Put a dot for the outliers.



Boxplot . . .

Draw the “whisker” from the quartiles to the observations that are lowest and highest within the inner fence!





Outliers

- In STATISTICS , an **outlier** is a data point that differs significantly from other observations.
- An outlier may be due to a variability in the measurement, an indication of novel data, or it may be the result of experimental error; the latter are sometimes excluded from the dataset.

Outlier

- An outlier can be an indication of exciting possibility, but can also cause serious problems in statistical analyses.
- Outliers can occur by chance in any distribution, but they can indicate novel behaviour or structures in the dataset, measurement error, or that the population has a heavy tailed distribution.

Outliers

- In the case of measurement error, one wishes to discard them or use statistics that are robust to outliers, while in the case of heavy-tailed distributions, they indicate that the distribution has high skewness and that one should be very cautious in using tools or intuitions that assume a normal distribution

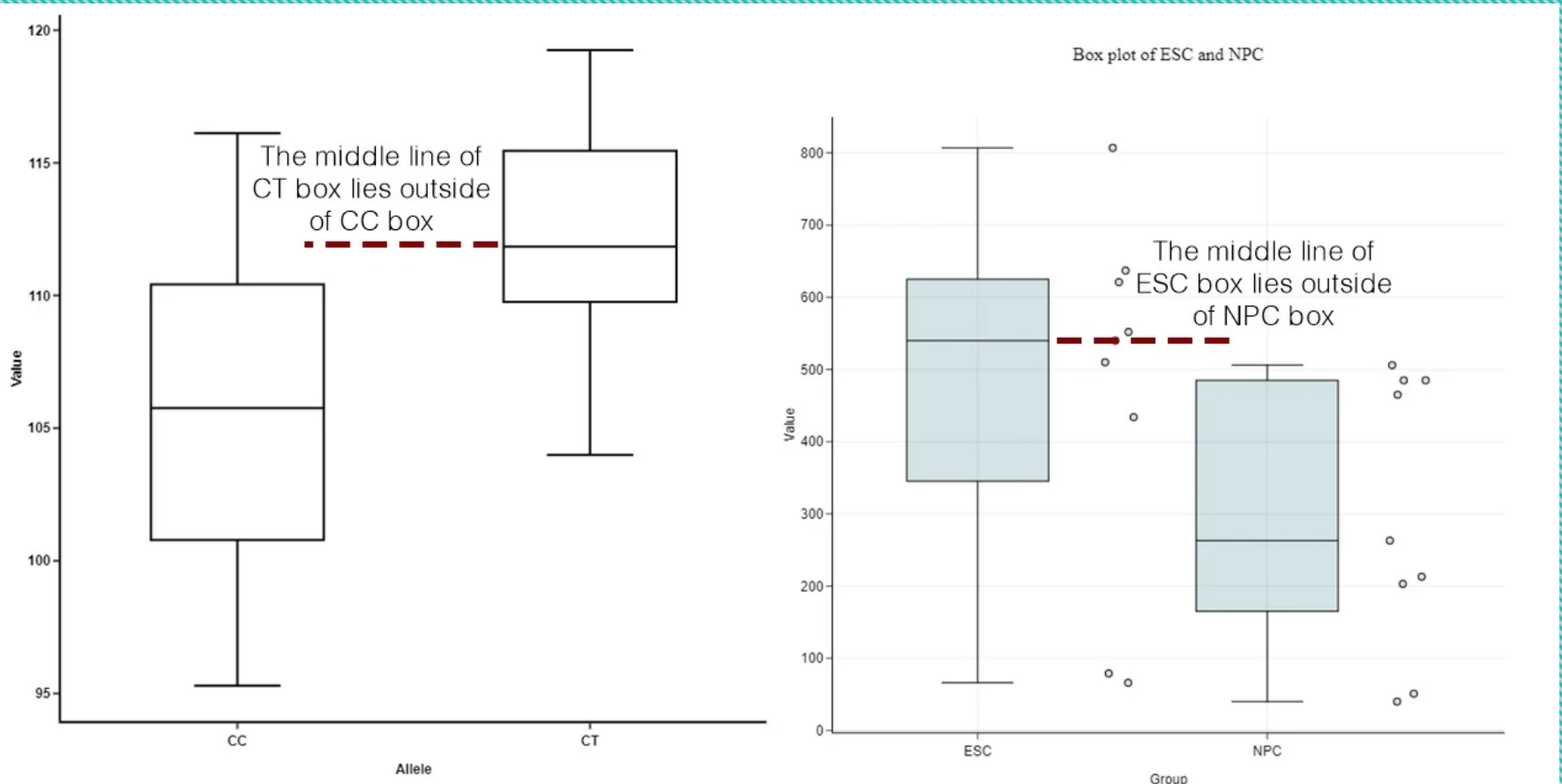
Outlier detection criteria

- A point beyond an inner fence on either side is considered a **mild outlier**.
- A point beyond an outer fence is considered an **extreme outlier**.

Example

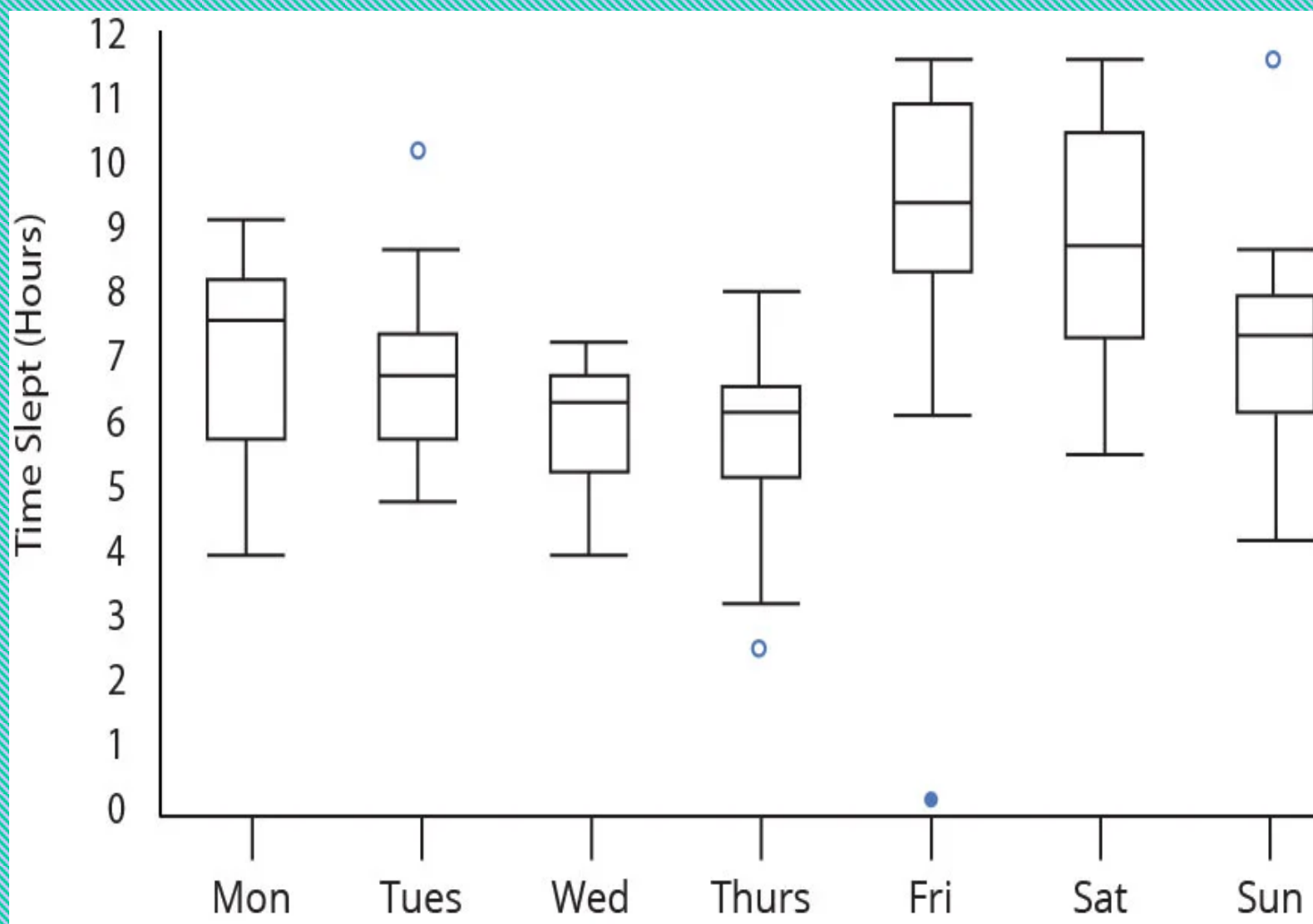
- The data set of $N = 90$ ordered observations as shown below is examined for outliers:
- 30, 171, 184, 201, 212, 250, 265, 270, 272, 289, 305, 306, 322, 322, 336, 346, 351, 370, 390, 404, 409, 411, 436, 437, 439, 441, 444, 448, 451, 453, 470, 480, 482, 487, 494, 495, 499, 503, 514, 521, 522, 527, 548, 550, 559, 560, 570, 572, 574, 578, 585, 592, 592, 607, 616, 618, 621, 629, 637, 638, 640, 656, 668, 707, 709, 719, 737, 739, 752, 758, 766, 792, 792, 794, 802, 818, 830, 832, 843, 858, 860, 869, 918, 925, 953, 991, 1000, 1005, 1068, 1441
- The computations are as follows:
- Median = $(n+1)/2$ largest data point = the average of the 45th and 46th ordered points = $(559 + 560)/2 = 559.5$
- $Q1 = .?$
- $Q3 = ?$

Compare the respective medians of each box plot. If the median line of a box plot lies outside of the box of a comparison box plot, then there is likely to be a difference between the two groups.

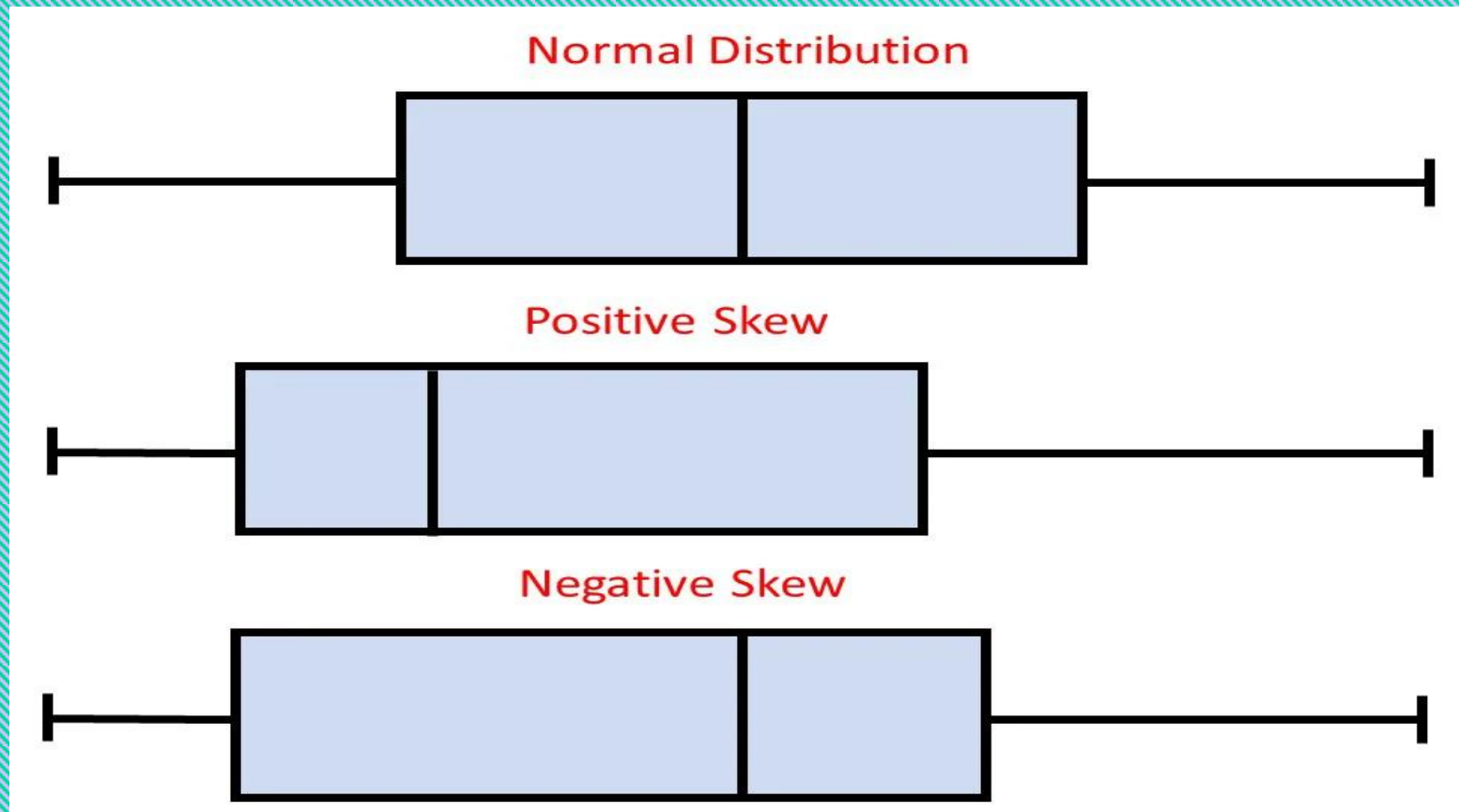


Compare the interquartile ranges and whiskers of box plots

- Compare the interquartile ranges (that is, the box lengths) to examine how the data is dispersed between each sample. The longer the box, the more dispersed the data. The smaller, the less dispersed the data.



Box plots are useful as they show the skewness of a data set



Skewness

- In statistical terms, **skewness** measures the asymmetry of the probability distribution of a real-valued random variable about its mean. In simpler terms, it quantifies how much a distribution deviates from a normal distribution. A distribution is **symmetrical** if it looks the same to the left and right of the center point.

Definition and Types of Skewness

- There are two main types of skewness — **left-skewed** (negatively skewed) and **right-skewed** (positively skewed). In a left-skewed distribution, the tail on the left side of the probability distribution is longer or fatter than the right. Conversely, the right tail is more prolonged or fatter in a right-skewed distribution.

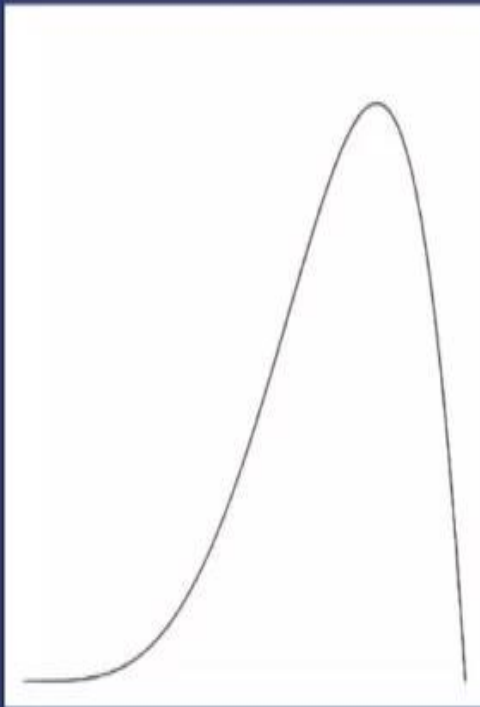
Why understanding Skewness is important

- Understanding skewness is critical for several reasons. First, it affects the choice of the appropriate statistics for data analysis.
- Second, skewness affects how we approach central tendency measures — the mean might not be the best representation of central tendency in skewed distributions.
- Third, knowing the type of skewness helps choose the proper statistical tests and understand the potential biases in data interpretation.

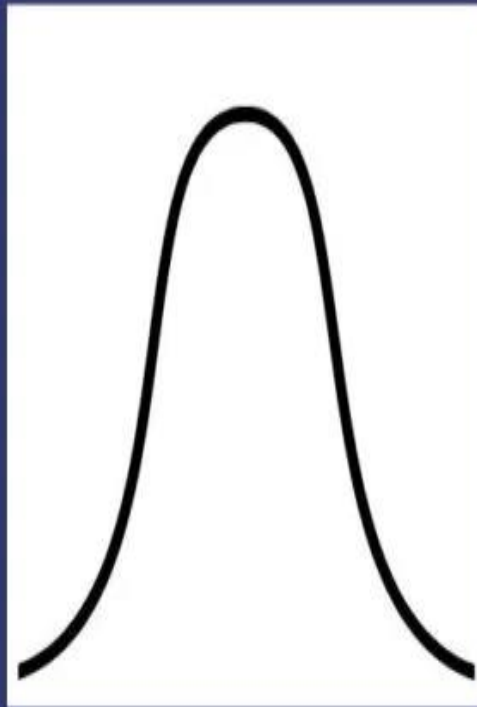
Some examples

- For example, right-skewed distributions are standard in financial data, which can impact investment strategies.
- Understanding the skewness of rainfall data in environmental science can be crucial for predicting extreme weather events.
- In medicine, skewed data can influence the interpretation of clinical trials and affect patient care decisions. Thus, skewness is not only about numbers; it's about grasping the underlying reality that data represents, aligning with the pursuit of accuracy and meaning in applying statistics.

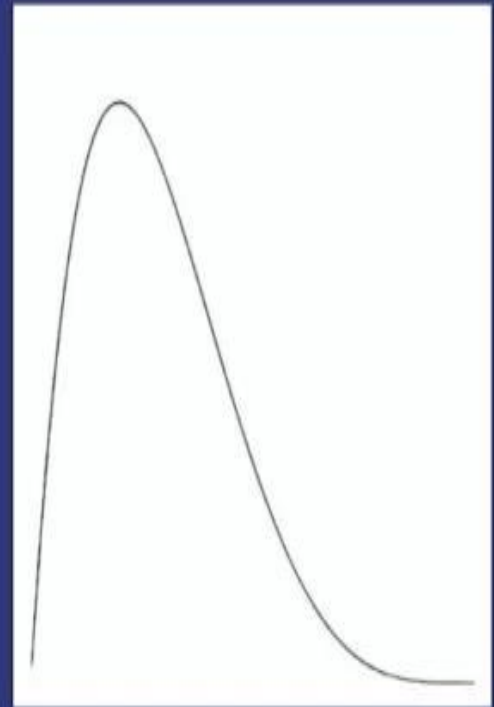
LEFT SKEWED
DISTRIBUTION



NO SKEW



RIGHT SKEWED
DISTRIBUTION



Boxplot using Python

- Code on next slide

Generate Random data

- `import matplotlib.pyplot as plt`
- `import numpy as np`
- `# Fixing random state for reproducibility`
- `np.random.seed(19680801)`
- `spread = np.random.rand(50) * 100`
- `center = np.ones(25) * 50`
- `flier_high = np.random.rand(10) * 100 + 100`
- `flier_low = np.random.rand(10) * -100`
- `data = np.concatenate((spread, center, flier_high, flier_low))`

How does `random.seed()` affect the randomness of generated numbers?

- *random.seed() affects the randomness by initializing the random number generator to a fixed state. This makes the sequence of numbers predictable and repeatable, but only if the same seed is used.*
- *Without setting a seed, the numbers generated will be different on each run.*

What happens if we don't use `random.seed()`?

- *If you don't use `random.seed()`, the random number generator will use the current system time or another source of entropy to initialize itself.*
- *This means you will get a different sequence of random numbers each time you run your program.*

Basic Box plot

```
fig, axs = plt.subplots(2, 3)
```

```
# basic plot
```

```
axs[0, 0].boxplot(data)
```

```
axs[0, 0].set_title('basic plot')
```

```
# notched plot
axs[0, 1].boxplot(data, 1)
axs[0, 1].set_title('notched plot')

# change outlier point symbols
axs[0, 2].boxplot(data, 0, 'gD')
axs[0, 2].set_title('change outlier\npoint symbols')

# don't show outlier points
axs[1, 0].boxplot(data, 0, "")
axs[1, 0].set_title("don't show\noutlier points")

# horizontal boxes
axs[1, 1].boxplot(data, 0, 'rs', 0)
axs[1, 1].set_title('horizontal boxes')

# change whisker length
axs[1, 2].boxplot(data, 0, 'rs', 0, 0.75)
axs[1, 2].set_title('change whisker length')

fig.subplots_adjust(left=0.08, right=0.98, bottom=0.05, top=0.9, hspace=0.4, wspace=0.3)
plt.show()
```